

Experiment No. 3

Title:

Implement control structures and form validation; create a grading system based on user-entered marks.

Software / Tools Required:

1. Visual Studio Code (VS Code)
2. Google Chrome
3. HTML5
4. JavaScript (ES6)

Theory:

Control structures are used in JavaScript to make decisions based on conditions. They allow different blocks of code to execute depending on user input.

(a) if Statement

Executes a block of code only if the given condition is true.

(b) if...else Statement

Executes one block if the condition is true, otherwise executes another block.

(c) else if Statement

Used to check multiple conditions one after another.

(d) switch Statement

Selects one block of code to execute from multiple possible cases.

(e) Form Validation

Form validation checks whether the user has entered valid input before processing the data. It helps prevent invalid or incomplete information.

Grading System

Accepts User Input Such As:

1. Student Name
2. Marks

It Displays:

1. Grade (A, B, C, D or Fail)
2. Validation Message
3. Error Message for Invalid Input

Result / Conclusion

The practical was completed successfully. The concepts of **control structures** and **form validation** were implemented. A grading system was developed that accepts user input, validates the entered marks, and displays the appropriate grade accurately using JavaScript.

Screenshot Attached